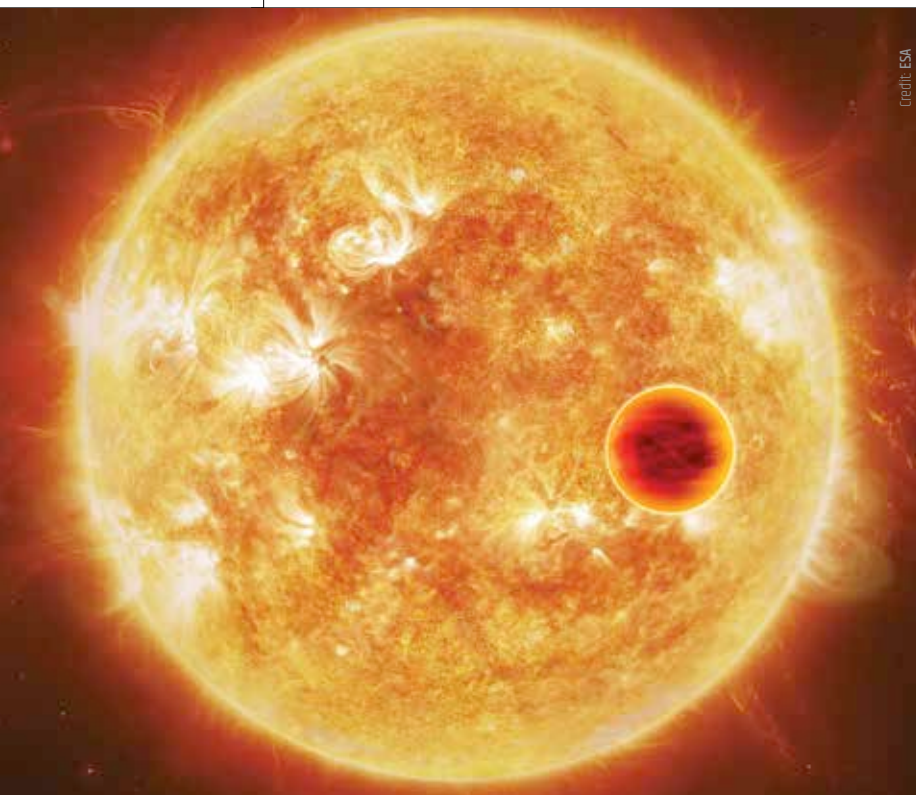


**OGO-1 satellite returns**

NASA's Orbiting Geophysics Observatory 1 (OGO-1), which was launched in 1964, is finally returning to Earth. <http://digit.in/sep20-67>

**Hubble maps giant halo**

NASA's Hubble has mapped the immense halo of gas enveloping the Andromeda galaxy. <http://digit.in/sep20-68>



Credit: ESA

TWINKLE

Studying the atmospheres of exoplanets

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Since the launch of the Kepler mission in 2009, the number of exoplanets we have found have increased significantly, with that mission alone identifying 2,347 exoplanets and many more candidate systems where exoplanets are expected to exist. Following up the work of the Kepler mission was the Transiting Exoplanet Survey Satellite or TESS mission, which is an ongoing complete sky survey that is expected to find at least 3000 more. We are just about

reaching a total of 4000 exoplanet candidates, with 2,757 confirmed identifications. Both these astronomical instruments look for the slight dipping in light as a planet passes in front of the host star, in the direction of Earth to identify exoplanets, called the transit method. As these instruments looked at only the exoplanets that passed in between the Earth and the host star, they found the systems in the sky where the orbital plane of the system is oriented towards the Earth.

The sheer number of exoplanets that we have found indicates that almost every star has a planet and the galaxy is simply teeming with potential habitats where life can take hold. The next step in identifying such habitats involves finding Earth-sized rocky planets in orbit around Sun like stars with an ozone

layer in the atmosphere for shielding the life from the ravages of UV light, and plenty of liquid water on the surface. In order to accomplish this, there was a need for an instrument that could analyse that little light that passes through the atmosphere of a planet while it crosses in front of the host star in relation to us. Chemicals and compounds in the atmosphere will absorb some of the light, and by carefully analysing these absorption lines, that look somewhat like barcodes, researchers can identify several characteristics of the planet such as its age, where in the system it was formed, if it is capable of supporting life, and if there are the chemicals associated with life in the atmosphere. This is where the Twinkle mission comes in, a relatively low cost instrument dedicated to following up the work of planet hunting telescopes, with studies of the atmospheres of transiting exoplanets. The Twinkle mission is a fast track mission that is being launched quickly to keep the work on the characterisation of exoplanets ongoing.

There are some ongoing and planned missions that have a similar task. The ongoing CHAracterising EXOPlanets Satellite (CHEOPS) mission is probing the atmospheres of super-Earths. The James Webb Space Telescope (JWST), or just called Webb, is the follow up mission to the Hubble Space Telescope that has a long list of science goals and may not be available to probe the atmospheres of candidate exoplanets where life is likely to exist. If Twinkle identifies another system with the telltale signs of life on any of its rocky exoplanets, then the JWST is likely to dedicate precious time to study that system. Then there is the PLANetary Transits and Oscillations of stars (PLATO) mission but that is scheduled for a launch in 2026. Most of the current and planned exoplanet surveys including TESS, Gaia, CHEOPS and